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FUJIWHARA on Five Centuries of Freezing Dates of Lake Suwa in the Central Japan.

By

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With 3 Figures.

Summary. As phenological indicators, the freezing dates of Lake Suwa in the Central Japan are tabulated in chronological order over more than five centuries since the winter 1443/44 up to the present. The years in the last $2^1/_2$ centuries have been characterized by warmer winter (i. e. later freezing), even though there is a definite sequence of fluctuation between cold and warm. Tree-rings analysis is also given for references.

Zusammenfassung. Die Gefriertermine des Suwa-Sees in Mitteljapan werden als phänologische Indikatoren in chronologischer Reihe über mehr als fünf Jahrhunderte, vom Winter 1443/44 bis heute, mitgeteilt. Die letzten $2^1/_2$ Jahrhunderte sind durch späteres Zufrieren, also wärmere Winter charakterisiert, wenn sich auch deutliche Fluktuationen zwischen kalt und warm erkennen lassen. Außerdem wird auf Jahresringen der Bäume beruhendes Beweismaterial diskutiert.

Résumé. L'auteur dispose en tableau chronologique les dates de gel du lac Suwa (Japon central) pour plus de 500 ans, soit depuis l'hiver 1443—44 jusqu'à aujourd'hui et qui peuvent servir d'indicateurs phénologiques. Les années des dernières $2^1/_2$ siècles sont caractérisées par des hivers plus chauds (gels plus tardifs), malgré des fluctuations définies de périodes chaudes et froides. Pour appuyer ces constations, l'auteur examine aussi les résultats obtenus par l'analyse des cernes des arbres.

The special purpose of this paper is to furnish the table of original data that has been compiled by the late Dr. S. FUJIWHARA (1884–1950), so that research workers of this interesting subject may have at hand the valuable material for further investigations. This is facilitated by the recent publication of FUJIWHARA's work [*1*] (edited by the writer), which concerns itself especially with the tabulation and graphical representation of freezing dates of Lake Suwa in the Central Japan.

During many years of study since 1912 [2], when his first study on the freezing dates of Lake Suwa was published, FUJIWHARA has gathered phenological material on the time of freezing of Lake Suwa, which embraces some 14.65 km² with a maximum depth of 7.6 m (Fig. 1). To assemble this information which related to the period extending from the distant past up to the present was not an easy task and required much time. The comparison and critical examination of text, the necessary selection and methodical classification of the material occupied his time for some forty years.

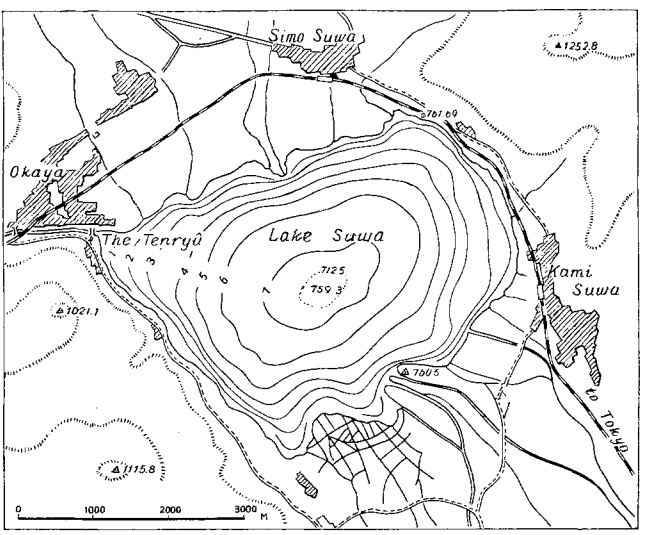


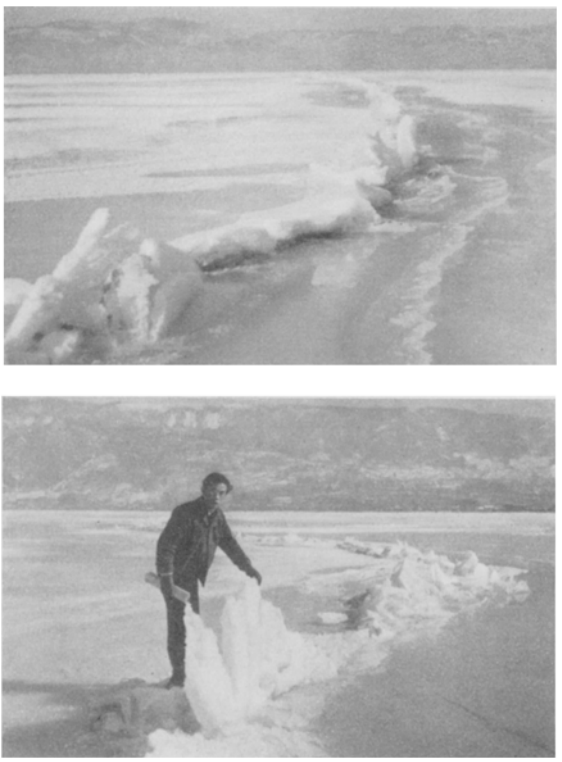
Fig. 1. Topography of the Area near Lake Suwa, Japan. Greatest depth of Lake Suwa: 7.6 m. (The two figures in the center of Fig. 1 should read 7.593 and 7.125, resp.)

In many cases it appears that phenological data are sometimes better for some plant-life comparisons than temperature records themselves, as the influence of temperature is "integrated", so to speak.

Study of the various data indicates that the winter freezing up is a good indicator of a "late", "early", "cold", or "warm" winter. The date of freezing is a function of temperature, not for any one day or even for several days, but for many days. Water has a high thermal capacity: hence, a protracted period is required in order to bring the water in condition for freezing. That the freezing dates give a fairly accurate record of winter severity in its own vicinity is evident.

The deformation of the ice sheet itself is a complicated process and the deformation as a result of the changes in the temperature of the ice occurs almost each winter. Whenever the temperature of the ice is increased or decreased, forces will be set up rendering to either expand or contract the ice. When the temperature of ice is lowered and if the ice is free to move, linear contraction sets in at the rate of about $^1/_{20\,000}$

for each degree of Centigrade. If the ice is not free to move the tensile stresses will accumulate until the limit of elasticity is reached, when relief will take place. These stresses resulting from a lowering temperature are usually a maximum at the surface and decrease with the increase in temperature as the surface of the ice (where the temperature is that of unfrozen water) is approached. That is to say, the tensile stresses are greatest at the surface and decrease toward the under surface of ice sheet.



a

b

Fig. 2 a and b. Characteristic rupture ("Omiwatari") taken on January 26, 1951. (Courtesy of the Suwa Weather Station.)

The tensile stresses in the horizontal directions accumulate until one or the other overcomes the strength of the ice and *cracks* form. As stated before, the tensile stresses are greatest at the surface and *rupture* will begin when these stresses overcome the strength of the ice at this place. When once started, the cracks will continue either partially or entirely through the thickness of the ice.

Cracks that are once formed often open again and again, but each time they reach under surface of the ice they fill with water. The water

rises about nine-tenths of the way to the surface and there solidifies. The diurnal and weekly change of temperature during the winter months cause a sufficient expansion and contraction of the ice covering Lake Suwa to shove up the rupture into pecular ridge, known as *"Omiwatari"*.

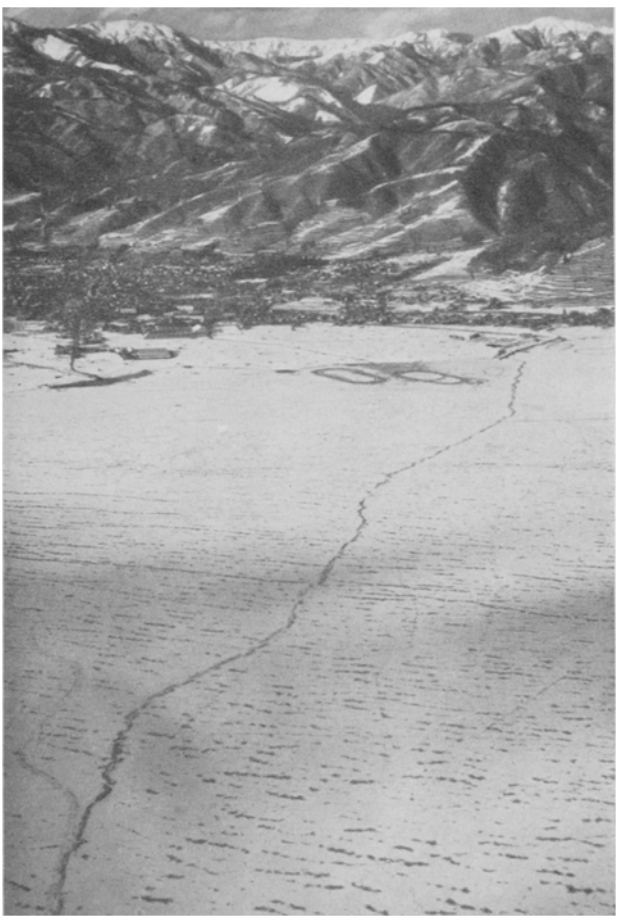


Fig. 3. Characteristic rupture ("Omiwatari") as seen from the aircraft, February 6, 1954. (Courtesy of the Asahi Press.)

These ridges are of various shapes and sizes, and are discontinuous of various length and range from a fraction of a meter to one meter in height. They are not ordinarily an accompaniment of very small or very large lakes, although their formation is not entirely controlled by the size of the lakes, depending in the region under discussion mainly upon the climatic conditions. They are best developed during seasons when the

Table I. *Freezing dates of Lake Suwa (Japan) compiled by* S. FUJIWHARA *(revised by* H. ARAKAWA*).*

XII: December; I: January; II: February; III: March; — no description; * approximate date, not later than; no: did not freeze or did not crack; did: freezed or cracked, missing dates; open: "Akiumi" (open sea), except land-fast ices which form along the shore; freezing date: full freez day.

Winter	Freezing dates	"Omiwatari"	Remarks	Winter	Freezing dates	"Omiwatari"	Remarks
1397–98	18 XII 1397	21 XII 1397		1480–81	3 I 1481	6 I 1481	
				81–82	15 I 1482	18 I 1482	
				82–83	24 I 1483	26 I 1483	
1443–44	7 I 1444	9 I 1444		83–84	31 XII 1483	1 I 1484	
44–45	23 XII 1444	26 XII 1444		84–85	1 I 1485	3 I 1485	
45–46	31 XII 1445	3 I 1446		85–86	10 I 1486	12 I 1486	
46–47	1 I 1447	4 I 1447		86–87	11 I 1487	14 I 1487	
47–48	30 XII 1447	2 I 1448		87–88	5 II 1488	7 II 1488	
48–49	7 I 1449	10 I 1449		88–89	20 I 1489	23 I 1489	
49–50	12 I 1450	15 I 1450		89–90	24 XII 1489	25 XII 1489	
1450–51	7 I 1451	11 I 1451		1490–91	9 II 1491	12 II 1491	
51–52	22 I 1452	25 I 1452		91–92	5 I 1492	7 I 1492	
52–53	28 XII 1452	31 XII 1452		92–93	1 I 1493	3 I 1493	
53–54	2 I 1454	4 I 1454		93–94	5 I 1494	7 I 1494	
54–55	4 I 1455	7 I 1455		94–95	8 I 1495	10 I 1495	
55–56	31 XII 1455	3 I 1456		95–96	3 I 1496	6 I 1496	
56–57	4 I 1457	7 I 1457		96–97	22 I 1497	25 I 1497	
57–58	no ?	no ?	open ?	97–98	11 XII 1497	13 XII 1497	
58–59	5 I 1459	8 I 1459		98–99	13 I 1499	14 I 1499	
59–60	20 XII 1459	23 XII 1459		99–1500	18 XII 1499	20 XII 1499	
1460–61	9 I 1461	12 I 1461		1500–01	27 XII 1500	29 XII 1500	
61–62	14 I 1462	17 I 1462		01–02	15 I 1502	18 I 1502	
62–63	29 XII 1462	31 XII 1462		02–03	11 I 1503	13 I 1503	
63–64	9 I 1464	12 I 1464		03–04	15 XII 1503	17 XII 1503	
64–65	10 I 1465	13 I 1465		04–05	31 XII 1504	2 I 1505	
65–66	11 XII 1465	14 XII 1465		05–06	no	no	open
66–67	20 I 1467	23 I 1467		06–07	20 XII 1506	22 XII 1506	
67–68	11 I 1468	13 I 1468		07–08	no	no	open
68–69	10 I 1469	12 I 1469		08–09	no	no	open
69–70	2 II 1470	5 II 1470		09–10	no	no	open
1470–71	26 XII 1470	30 XII 1470		1510–11	no	no	open
71–72	22 XII 1471	26 XII 1471		11–12	no	no	open
72–73	24 XII 1472	27 XII 1472		12–13	no	no	open
73–74	3 I 1474	7 I 1474		13–14	no	no	open
74–75	1 I 1475	4 I 1475		14–15	no	no	open
75–76	31 XII 1475	3 I 1476		15–16	1 I 1516	3 I 1516	
76–77	6 I 1477	8 I 1477		16–17	5 I 1517	8 I 1517	
77–78	17 I 1478	19 I 1478		17–18	5 II 1518	8 II 1518	
78–79	16 I 1479	19 I 1479		18–19	29 XII 1518	31 XII 1518	
79–80	5 I 1480	8 I 1480		19–20	3 I 1520	5 I 1520	

(Table I continued.)

Winter	Freezing dates	"Omiwatari"	Remarks
1520–21	14 I 1521	17 I 1521	
21–22	26 I 1522	28 I 1522	
22–23	1 I 1523	4 I 1523	
23–24	1 I 1524	4 I 1524	
24–25	11 I 1525	14 I 1525	
25–26	14 XII 1525	16 XII 1525	
26–27	14 I 1527	16 I 1527	
27–28	16 I 1528	19 I 1528	
28–29	27 XII 1528	9 I 1529	
29–30	6 I 1530	8 I 1530	
1530–31	1 I 1531	4 I 1531	
31–32	2 I 1532	5 I 1532	
32–33	9 I 1533	11 I 1533	
33–34	21 XII 1533	24 XII 1533	
34–35	16 I 1535	18 I 1535	
35–36	21 XII 1535	23 XII 1535	
36–37	5 II 1537	8 II 1537	
37–38	13 I 1538	16 I 1538	
38–39	18 XII 1538	20 XII 1538	
39–40	13 II 1540	15 II 1540	
1540–41	23 XII 1540	26 XII 1540	
41–42	3 I 1542	5 I 1542	
42–43	15 I 1543	17 I 1543	
43–44	4 I 1544	6 I 1544	
44–45	13 I 1545	15 I 1545	
45–46	1 II 1546	3 II 1546	
46–47	no	no	open
47–48	5 I 1548	7 I 1548	
48–49	13 I 1549	15 I 1549	
49–50	7 I 1550	9 I 1550	
1550–51	4 I 1551	6 I 1551	
51–52	3 II 1552	5 II 1552	
52–53	25 XII 1552	27 XII 1552	
53–54	did	did	
54–55	8 I 1555	10 I 1555	
55–56	no	no	open
56–57	2 I 1557	4 I 1557	
57–58	24 I 1558	26 I 1558	
58–59	31 XII 1558	2 I 1559	
59–60	2 I 1560	4 I 1560	
1560–61	5 II 1561	7 II 1561	
61–62	10 I 1562	12 I 1562	
62–63	4 I 1563	6 I 1563	
63–64	12 I 1564	14 I 1564	
64–65	6 I 1565	8 I 1565	
65–66	30 XII 1565	1 I 1566	
66–67	8 I 1567	14 I 1567	
67–68	4 I 1568	6 I 1568	
68–69	25 XII 1568	28 XII 1568	
69–70	22 XII 1569	24 XII 1569	
1570–71	26 I 1571	28 I 1571	
71–72	22 XII 1571	25 XII 1571	
72–73	16 I 1573	18 I 1573	
73–74	30 XII 1573	1 I 1574	
74–75	7 II 1575	9 II 1575	
75–76	25 I 1576	28 I 1576	
76–77	1 I 1577	4 I 1577	
77–78	5 I 1578	7 I 1578	
78–79	28 I 1579	30 I 1579	
79–80	13 I 1580	16 I 1580	
1580–81	11 I 1581	13 I 1581	
81–82	29 I 1582	31 I 1582	
82–83	17 XII 1582	19 XII 1582	
83–84	18 XII 1583	20 XII 1583	
84–85	28 XII 1584	31 XII 1584	
85–86	14 I 1586	16 I 1586	
86–87	28 XII 1586	30 XII 1586	
87–88	23 XII 1587	26 XII 1587	
88–89	14 I 1589	16 I 1589	
89–90	31 XII 1589	2 I 1590	
1590–91	13 I 1591	15 I 1591	
91–92	13 I 1592	16 I 1592	
92–93	28 XII 1592	30 XII 1592	
93–94	4 III 1594	6 III 1594	
94–95	—	—	—
95–96	24 I 1596	26 I 1596	
96–97	25 I 1597	27 I 1597	
97–98	13 XII 1597	16 XII 1597	
98–99	20 XII 1598	23 XII 1598	
99–1600	21 I 1600	24 I 1600	
1600–01	19 XII 1600	22 XII 1600	
01–02	29 I 1602	1 II 1602	
02–03	3 I 1603	6 I 1603	
03–04	31 XII 1603	2 I 1604	
04–05	14 I 1605	18 I 1605	
05–06	18 XII 1605	21 XII 1605	
06–07	23 XII 1606	25 XII 1606	
07–08	24 I 1608	27 I 1608	
08–09	5 I 1609	9 I 1609	
09–10	18 XII 1609	21 XII 1609	
1610–11	31 XII 1610	3 I 1611	
11–12	22 XII 1611	25 XII 1611	
12–13	25 I 1613	28 I 1613	
13–14	9 I 1614	11 I 1614	
14–15	21 XII 1614	24 XII 1614	
15–16	22 I 1616	25 I 1616	
16–17	22 XII 1616	25 XII 1616	
17–18	2 I 1618	5 I 1618	
18–19	10 I 1619	12 I 1619	
19–20	10 I 1620	14 I 1620	

(Table I continued.)

Winter	Freezing dates	"Omiwatari"	Remarks
1620–21	8 XII 1620	11 XII 1620	
21–22	7 I 1622	10 I 1622	
22–23	12 I 1623	14 I 1623	
23–24	24 I 1624	26 I 1624	
24–25	22 I 1625	24 I 1625	
25–26	8 I 1626	11 I 1626	
26–27	9 I 1627	12 I 1627	
27–28	8 I 1628	31 I 1628	
28–29	21 XII 1628	23 XII 1628	
29–30	20 I 1630	22 I 1630	
1630–31	22 XII 1630	24 XII 1630	
31–32	24 I 1632	26 I 1632	
32–33	18 I 1633	21 I 1633	
33–34	9 I 1634	12 I 1634	
34–35	28 XII 1634	30 XII 1634	
35–36	2 I 1636	4 I 1636	
36–37	11 I 1637	14 I 1637	
37–38	31 XII 1637	2 I 1638	
38–39	21 I 1639	23 I 1639	
39–40	19 I 1640	21 I 1640	
1640–41	18 I 1641	20 I 1641	
41–42	15 XII 1641	17 XII 1641	
42–43	25 I 1643	27 I 1643	
43–44	5 I 1644	7 I 1644	
44–45	1 I 1645	3 I 1645	
45–46	12 I 1646	15 I 1646	
46–47	9 II 1647	11 II 1647	
47–48	4 I 1648	6 I 1648	
48–49	29 XII 1648	1 I 1649	
49–50	5 I 1650	7 I 1650	
1650–51	14 I 1651	16 I 1651	
51–52	13 I 1652	15 I 1652	
52–53	6 I 1653	8 I 1653	
53–54	13 I 1654	15 I 1654	
54–55	7 I 1655	9 I 1655	
55–56	10 I 1656	12 I 1656	
56–57	10 I 1657	12 I 1657	
57–58	24 I 1658	26 I 1658	
58–59	28 XII 1658	30 XII 1658	
59–60	10 I 1660	12 I 1660	
1660–61	11 I 1661	13 I 1661	
61–62	18 I 1662	20 I 1662	
62–63	1 I 1663	4 I 1663	
63–64	10 I 1664	12 I 1664	
64–65	26 XII 1664	28 XII 1664	
65–66	28 XII 1665	30 XII 1665	
66–67	5 I 1667	8 I 1667	
67–68	10 I 1668	14 I 1668	
68–69	16 XII 1668	19 XII 1668	
69–70	25 I 1670	28 I 1670	
1670–71	6 I 1671	9 I 1671	
71–72	no	no	open
72–73	26 XII 1672	28 XII 1672	
73–74	22 I 1674	24 I 1674	
74–75	18 I 1675	20 I 1675	
75–76	21 I 1676	23 I 1676	
76–77	5 I 1677	7 I 1677	
77–78	2 II 1678	4 II 1678	
78–79	5 I 1679	7 I 1679	
79–80	28 XII 1679	30 XII 1679	
1680–81	28 XII 1680	1 I 1681	
81–82	21 I 1682	23 I 1682	
82–83	—	—	—
83–84	did	6 I 1684*	
84–85	did	10 I 1685*	
85–86	did	11 I 1686*	
86–87	did	6 I 1687*	
87–88	did	12 I 1688*	
88–89	did	17 I 1689*	
89–90	did	6 I 1690*	
1690–91	did	7 I 1691*	
91–92	did	21 I 1692*	
92–93	did	29 XII 1692*	
93–94	did	9 I 1694*	
94–95	did	11 I 1695*	
95–96	did	9 I 1696*	
96–97	29 XII 1696	1 I 1697	
97–98	did	2 I 1698*	
98–99	did	8 I 1699*	
99–1700	did	4 I 1700*	
1700–01	did	20 I 1701*	
01–02	did	11 II 1702*	
02–03	no	no	open
03–04	did	8 II 1704*	
04–05	did	2 III 1705*	
05–06	did	27 I 1706*	
06–07	did	2 I 1707*	
07–08	did	6 I 1708*	
08–09	did	9 I 1709*	
09–10	did	6 II 1710*	
1710–11	—	no	
11–12	did	8 I 1712*	
12–13	did	7 I 1713*	
13–14	did	29 XII 1713*	
14–15	did	30 I 1715*	
15–16	did	19 I 1716*	
16–17	did	28 I 1717*	
17–18	did	28 I 1718*	
18–19	did	18 I 1719*	
19–20	did	14 I 1720*	

(Table I continued.)

Winter	Freezing dates	"Omiwatari"	Remarks
1720–21	did	25 I 1721*	
21–22	did	7 II 1722*	
22–23	did	29 I 1723*	
23–24	did	6 I 1724*	
24–25	no	no	open
25–26	did	10 I 1726*	
26–27	did	4 I 1727*	
27–28	did	30 I 1728*	
28–29	did	25 I 1729*	
29–30	did	17 I 1730*	
1730–31	did	3 I 1731*	
31–32	did	28 I 1732*	
32–33	did	10 I 1733*	
33–34	did	1 II 1734*	
34–35	did	7 I 1735*	
35–36	did	6 I 1736*	
36–37	did	11 I 1737*	
37–38	no	no	open
38–39	did	6 I 1739*	
39–40	did	no	nearly open
1740–41	did	18 I 1741*	
41–42	16 I 1742	18 I 1742	
42–43	did	6 I 1743*	
43–44	did	6 II 1744*	
44–45	did	31 XII 1744*	
45–46	did	no	nearly open
46–47	8 I 1747	13 I 1747	
47–48	27 I 1748	1 II 1748	
48–49	13 I 1749	18 I 1749	
49–50	23 I 1750	28 I 1750	
1750–51	2 I 1751	7 I 1751	
51–52	29 XII 1751	3 I 1752	
52–53	7 I 1753	13 I 1753	
53–54	did	26 I 1754*	
54–55	1 II 1755	7 II 1755	
55–56	1 I 1756	11 I 1756	
56–57	5 I 1757	12 I 1757	
57–58	did	22 I 1758*	
58–59	16 I 1759	25 I 1759	
59–60	no	no	open
1760–61	31 XII 1760	5 I 1761	
61–62	26 XII 1761	7 I 1762	
62–63	12 I 1763	20 I 1763	
63–64	did	14 I 1764*	
64–65	did	11 I 1765*	
65–66	did	24 I 1766*	
66–67	did	18 I 1767*	
67–68	did	19 I 1768*	
68–69	18 I 1769	27 I 1769	
69–70	9 I 1770	14 I 1770	
1770–71	no	no	open
71–72	did	did	
72–73	did	18 I 1773*	
73–74	did	23 I 1774*	
74–75	29 XII 1774	5 I 1775	
75–76	28 I 1776	4 II 1776	
76–77	did	21 I 1777*	
77–78	did	13 I 1778*	
78–79	did	9 II 1779*	
79–80	did	17 I 1780*	
1780–81	did	17 I 1781*	
81–82	did	26 I 1782*	
82–83	did	22 I 1783*	
83–84	did	18 I 1784*	
84–85	did	12 XII 1784*	
85–86	did	22 I 1786*	
86–87	did	17 I 1787*	
87–88	did	4 I 1788*	
88–89	1 I 1789	11 I 1789	
89–90	10 I 1790	15 I 1790	
1790–91	30 XII 1790	8 I 1791	
91–92	did	did	
92–93	28 XII 1792	9 I 1793	
93–94	no	no	open
94–95	25 I 1795	4 II 1795	
95–96	did	19 I 1796*	
96–97	2 I 1797	6 I 1797	
97–98	6 I 1798	11 I 1798	
98–99	10 I 1799	20 I 1799	
99–1800	no	no	open
1800–01	6 I 1801	12 I 1801	
01–02	did	did	
02–03	no	no	nearly open
03–04	17 I 1804	27 I 1804	
04–05	no	no	open
05–06	3 II 1806	12 II 1806	
06–07	5 I 1807	13 I 1807	
07–08	11 I 1808	22 I 1808	
08–09	7 I 1809	17 I 1809	
09–10	27 XII 1809	6 I 1810	
1810–11	4 I 1811	14 I 1811	
11–12	4 I 1812	10 I 1812	
12–13	26 XII 1812	30 XII 1812	
13–14	13 I 1814	18 I 1814	
14–15	did	8 I 1815*	
15–16	31 XII 1815	3 I 1816	
16–17	21 I 1817	23 I 1817	
17–18	15 I 1818	17 I 1818	
18–19	did	1 II 1819*	
19–20	16 I 1820	17 I 1820	

(Table I continued.)

Winter	Freezing dates	"Omiwatari"	Remarks
1820–21	6 I 1821	7 I 1821	
21–22	10 I 1822	12 I 1822	
22–23	22 I 1823	23 I 1823	
23–24	did	did	
24–25	no ?	no ?	open ?
25–26	did	did	
26–27	did	did	
27–28	did	did	
28–29	8 I 1829	21 I 1829	
29–30	16 I 1830	19 I 1830	
1830–31	13 I 1831	16 I 1831	
31–32	9 I 1832	11 I 1832	
32–33	did	18 I 1833*	
33–34	7 II 1834	10 II 1834	
34–35	13 I 1835	14 I 1835	
35–36	5 I 1836	9 I 1836	
36–37	11 I 1837	13 I 1837	
37–38	6 II 1838	8 II 1838	
38–39	28 I 1839	29 I 1839	
39–40	13 I 1840	14 I 1840	
1840–41	24 I 1841	27 I 1841	
41–42	7 I 1842	10 I 1842	
42–43	no	no	open
43–44	17 I 1844	19 I 1844	
44–45	no	no	open
45–46	28 XII 1845	2 I 1846	
46–47	10 I 1847	16 I 1847	
47–48	did	14 II 1848*	
48–49	did	4 II 1849*	
49–50	did	27 I 1850*	
1850–51	5 II 1851	8 II 1851	
51–52	did	30 I 1852*	
52–53	2 II 1853	6 II 1853	
53–54	did	no	nearly open
54–55	6 II 1855	11 II 1855	
55–56	12 I 1856	15 I 1856	
56–57	did	6 II 1857*	
57–58	8 II 1858	11 II 1858	
58–59	23 I 1859	28 I 1859	
59–60	did	27 I 1860*	
1860–61	30 I 1861	2 II 1861	
61–62	did	6 II 1862*	
62–63	21 I 1863	23 I 1863	
63–64	17 I 1864	18 I 1864	
64–65	—	—	—
65–66	no	no	open
66–67	no	no	open
67–68	no	no	open
68–69	no	no	open
69–70	21 I 1870	22 I 1870	
1870–71	23 I 1871	25 I 1871	
71–72	18 I 1872	26 I 1872	
72–73	did	did	
73–74	did	did	
74–75	did	did	
75–76	did	did	
76–77	did	did	
77–78	did	did	
78–79	did	did	
79–80	no	no	open
1880–81	did	did	
81–82	did	did	
82–83	did	did	
83–84	did	did	
84–85	did	did	
85–86	did	did	
86–87	did	did	
87–88	did	did	
88–89	did	did	
89–90	no	no	open
1890–91	did	did	
91–92	did	did	
92–93	did	21 I 1893*	
93–94	did	30 I 1894*	
94–95	did	28 I 1895*	
95–96	did	14 II 1896*	
96–97	did	27 I 1897*	
97–98	26 XII 1897	19 I 1898*	
98–99	18 I 1899	28 I 1899*	
99–1900	9 I 1900	8 II 1900*	
1900–01	19 I 1901	30 I 1901*	
01–02	31 XII 1901	26 I 1902*	
02–03	15 I 1903	16 II 1903*	
03–04	22 XII 1903	9 I 1904*	
04–05	25 XII 1904	12 I 1905*	
05–06	16 I 1906	29 I 1906*	
06–07	30 XII 1906	27 I 1907*	
07–08	24 XII 1907	18 I 1908*	
08–09	2 I 1909	24 I 1909*	
09–10	27 XII 1909	15 II 1910*	
1910–11	3 I 1911	23 I 1911*	
11–12	4 I 1912	8 II 1912*	
12–13	2 I 1913	30 I 1913*	
13–14	8 I 1914	no	nearly open
14–15	10 I 1915	24 I 1915*	
15–16	no	no	open
16–17	did	23 II 1917*	
17–18	23 XII 1917	31 I 1918*	
18–19	6 I 1919	9 II 1919*	
19–20	26 I 1920	27 II 1920*	

(Table I continued.)

Winter	Freezing dates	"Omiwatari"	Remarks	Winter	Freezing dates	"Omiwatari"	Remarks
1920–21	8 I 1921	5 II 1921*		1938–39	4 I 1939	8 I 1939	
21–22	5 I 1922	27 I 1922*		39–40	17 I 1940	22 I 1940	
22–23	20 XII 1922	24 I 1923*					
23–24	24 I 1924	30 I 1924		1940–41	2 II 1941	5 II 1941	
24–25	15 I 1925	17 I 1925		41–42	20 I 1942	22 I 1942	
25–26	3 I 1926	7 I 1926		42–43	6 I 1943	8 I 1943	
26–27	26 XII 1926	28 XII 1926		43–44	9 I 1944	19 I 1944	
27–28	4 I 1928	8 I 1928		44–45	3 I 1945	5 I 1945	
28–29	31 XII 1928	5 I 1929		45–46	27 XII 1945	3 I 1946	
29–30	9 I 1930	11 I 1930		46–47	4 I 1947	8 I 1947	
				47–48	20 XII 1947	24 XII 1947	
1930–31	11 I 1931	18 I 1931		48–49	no	no	open
31–32	no	no	open	49–50	12 I 1950	13 I 1950	
32–33	14 I 1933	16 I 1933					
33–34	6 I 1934	10 I 1934		1950–51	2 I 1951	9 I 1951	
34–35	7 II 1935	9 II 1935		51–52	13 I 1952	19 I 1952	
35–36	21 XII 1935	31 XII 1935		52–53	5 I 1953	no	
36–37	13 I 1937	no	nearly open	53–54	25 I 1954	27 I 1954	
37–38	4 I 1938	9 I 1938					

The periods 1444–1682 and after 1923 to the present include the observations, relatively homogeneous among themselves and having a sufficient degree of accuracy.

The period 1682–1923 contains the data, which are much inferior to modern or older observations and are regarded only as supplementary, but still they can serve.

For example, we find for 1443/44 winter noted as follows: The freezing date, January 7, 1444 and "Omiwatari" date, January 9, 1444.

This list shows at a glance the approximate character of all winters for a period of five centuries, except only few winters (1457/8, 1594/5, 1682/3, 1824/5, and 1864/5). For an easier survey, the following gives the absolute extremes of the freezing dates.

	Freezing	"Omiwatari"
Most early date ..	8 XII 1620	11 XII 1620
Latest date	4 III 1594	6 III 1594

snowfall is very light. In the cold winter 1951/52, they are poorly developed on Lake Suwa, for the snowfall was unusually heavy (exceeding 10 cm in depth). They do occur on the shallow lakes such as Lake Suwa, for such admit a heavy coating of ice. The characteristic ruptures are formed again and again, and henceforward we shall reserve the term "Omiwatari" for the first "Omiwatari" so defined that is formed at first.

Freezing dates of Lake Suwa have been kept by Suwa shrine, YATSURUGI temple, Mr. Y. SUWA and others. More recent freezing dates were kept at the Suwa weather station.

The characteristic ruptures or "Omiwatari" occurring usually two or three days after the full freeze of Lake Suwa which was covered with ice are shown in Fig. 2*a*, 2*b* and 3.

To show climatic history, sequences of freezing dates of Lake Suwa may be tabulated as a chronological table. The dates of freezing and "Omiwatari" for Lake Suwa are given in Table I. Most of these data have been published in Japanese by Viscount A. TANAKA [*4*] who treated this matter only partially. The revision and discussion of the historical data necessitated a complete rehandling of this existing text. We are convinced that these historical data extending over more than 5 centuries are often remarkably exact; they constitute a unique and valuable source of climatology.

Thus the freezing dates of Lake Suwa as phenological indicator immediately provide the source for a 500-year chronology of winter temperature. From the last compilation I make the following Table II. In it the period 1443/44–1953/54 is divided into two about $2^1/_2$ century periods, and the frequencies of "Omiwatari" dates are given for each decade 11 XII/20 XII, 21 XII/30 XII, 31 XII/9 I, 10 I/19 I, 20 I/29 I, 30 I/8 II, 9 II/18 II and 19 II–. Taken together these two about $2^1/_2$ century periods show the following frequency of cold and mild winters (percents in parentheses).

Table II. *Frequency of cold and mild winters in two $2^1/_2$ century periods since 1443/44 up to the present.*

Decade	The period 1443/1444—1699/1700		The period after 1700/1701 to the present	
11 XII–20 XII	12	(5%)	1	(—)
21 XII–30 XII	36	(15%)	4	(2%)
31 XII– 9 I	81	(**33%**)	40	(20%)
10 I –19 I	65	(27%)	**61**	(**31%**)
20 I –29 I	33	(14%)	50	(25%)
30 I – 8 II	11	(4%)	30	(15%)
9 II –18 II	4	(2%)	12	(6%)
19 II –	1	(—)	2	(1%)
Total...	243		200	

The maximum frequencies are printed in bold faced type. The 250-year period 1444/1700 was apparently lower in winter temperature than similar interval in the last $2^1/_2$ centuries. The effect on the freezing dates of Lake Suwa was decisive.

In this period of 511 years we count 40 "open winters" without full freeze except land-fast ices.

Table III. *An index of the more mild winters in which Lake Suwa remained nearly open from 1444 to 1954.*

15th century	1457–58 ?	13
16th century	1505–06, 1507–08, 1508–09, 1509–10, 1510–11, 1511–12, 1512–13, 1513–14, 1514–15, 1546–47, 1555–56	
17th century	1671–72	
18th century	1702–03, 1710–11, 1724–25, 1737–38, 1739–40, 1745–46, 1759–60, 1770–71, 1793–94	27
19th century	1799–1800, 1802–03, 1804–05, 1824–25 ?, 1842–43, 1844–45, 1853–54, 1865–66, 1866–67, 1867–68, 1868–69, 1879–80, 1889–90	
20th century	1913–14, 1915–16, 1931–32, 1936–37, 1948–49	

The early records of Lake Suwa clearly show a long period (1450–1700) much colder than the average. The years since have been characterized by warmer winter (i. e. later freezing) even though there is a definite sequence of fluctuation between cold and warm. The frequency of mild winters in the last $2^1/_2$ centuries is a striking phenomenon that it has been noticed by many climatologists.

The outstanding winters in Japan for a period of five centuries, in the sense of very hard, were tabulated in Table IV. Early full freezing before 20th of December, and therefore early winters are:

Table IV. *An index of the more hard winters with early freezing dates.*

Winter	Freezing dates	"Omiwatari"
1465–1466	11 XII 1465	14 XII 1465
1497–1498	11 XII 1497	13 XII 1497
1499–1500	18 XII 1499	20 XII 1499
1503–1504	15 XII 1503	17 XII 1503
1525–1526	14 XII 1525	16 XII 1525
1538–1539	18 XII 1538	20 XII 1538
1582–1583	17 XII 1582	19 XII 1582
1583–1584	18 XII 1583	20 XII 1583
1597–1598	13 XII 1597	16 XII 1597
1620–1621	8 XII 1620	11 XII 1620
1641–1642	15 XII 1641	17 XII 1641
1668–1669	16 XII 1668	19 XII 1668
1784–1785	did	12 XII 1784

It is clear that early years of records were distinctly colder than more recent ones.

Table V. *Annual ring-widths of the long-lived Japanese cypress of central Japan in millimeters (from* K. YAMAZAWA*).*

Year	0	1	2	3	4	5	6	7	8	9
1110	—	—	—	—	—	—	—	—	—	0.6
20	0.4	0.4	0.3	0.6	0.4	0.2	0.5	0.6	0.6	0.7
30	0.9	0.7	1.0	0.6	1.1	0.6	0.9	0.8	0.7	0.6
40	1.0	0.6	0.5	0.6	0.9	0.9	0.8	0.8	0.7	0.6
1150	0.6	0.6	0.9	0.8	0.6	0.6	0.6	0.4	0.6	0.6
60	1.0	0.7	0.7	0.8	0.9	0.9	0.9	0.9	0.9	0.8
70	0.8	0.9	0.9	0.8	0.9	0.9	1.0	1.0	0.9	0.9
80	0.8	0.8	1.0	0.8	1.3	1.1	0.8	0.9	1.4	0.9
90	1.2	0.9	0.9	1.1	1.2	1.4	1.3	1.4	1.3	0.9
1200	1.0	1.2	1.0	1.2	0.9	0.9	1.2	0.9	1.0	1.2
10	1.0	1.1	0.6	0.7	0.8	0.7	0.6	0.55	0.60	0.60
20	0.65	0.60	0.75	0.80	0.90	0.90	0.90	0.90	0.95	0.95
30	0.95	0.90	1.00	1.10	0.95	0.95	0.95	0.90	0.70	0.90
40	0.70	0.90	0.85	0.90	0.90	0.95	0.90	1.05	1.15	1.05
1250	1.15	0.80	0.85	1.00	0.90	0.90	0.85	0.85	0.85	1.20
60	1.00	1.00	0.75	0.90	1.00	1.15	0.95	1.00	0.85	0.75
70	1.20	0.85	1.25	1.10	1.15	1.00	0.90	0.85	0.75	0.70
80	0.55	0.80	0.75	0.80	0.80	0.70	0.70	0.95	0.85	0.80
90	0.80	0.70	0.70	0.75	0.65	0.65	0.55	0.43	0.50	0.37
1300	0.33	0.27	0.30	0.43	0.43	0.47	0.47	0.27	0.40	0.43
10	0.27	0.20	0.27	0.33	0.67	1.07	1.20	1.33	1.67	1.57
20	1.53	1.38	1.95	1.45	1.73	1.55	1.58	1.35	1.28	1.75
30	1.18	1.53	1.18	1.43	1.43	1.10	0.98	0.85	1.20	1.70
40	1.50	1.25	1.10	1.23	1.23	1.30	1.10	1.00	1.40	1.40
1350	1.65	1.53	1.45	1.13	1.20	1.63	1.60	0.95	1.55	1.68
60	1.85	1.58	1.75	1.90	1.33	1.08	0.90	0.83	0.98	1.03
70	1.05	1.25	1.55	1.55	1.50	1.38	1.25	1.45	1.43	1.40
80	1.40	1.50	1.40	1.68	1.45	1.98	2.00	1.83	1.35	1.25
90	1.63	1.53	1.48	1.35	1.53	1.63	1.63	1.30	1.08	1.23
1400	1.55	1.23	1.63	2.10	1.60	1.48	1.38	1.50	1.55	1.65
10	1.90	1.65	1.50	1.48	1.68	1.48	1.50	1.45	1.23	1.30
20	1.28	1.23	1.13	1.08	1.35	1.73	1.63	1.40	1.00	1.05
30	1.23	0.93	2.10	1.90	1.85	1.40	1.25	1.23	1.18	1.33
40	1.13	1.45	0.95	1.18	1.73	1.08	1.33	1.18	1.20	1.03
1450	0.95	1.05	1.53	1.35	1.73	1.15	1.23	0.93	0.93	1.23
60	1.50	1.28	1.28	1.15	0.98	0.80	0.75	1.43	1.20	1.08
70	0.88	0.80	0.83	0.95	0.88	0.90	0.93	0.93	0.90	1.13
80	1.03	0.98	0.90	0.83	0.85	1.13	1.05	0.90	1.13	1.13
90	1.30	1.20	1.25	1.08	1.25	1.55	0.98	1.23	1.78	1.68
1500	1.40	1.38	1.18	1.38	1.48	0.70	0.95	1.33	1.15	0.98
10	1.13	1.25	1.80	1.43	2.00	2.05	2.05	1.63	1.38	1.55
20	1.15	1.25	2.08	1.80	1.35	1.35	1.50	1.43	1.25	1.48
30	1.68	1.43	1.35	1.25	1.10	0.90	1.28	0.95	0.93	0.90
40	1.30	1.58	1.43	1.63	1.30	1.15	1.03	0.80	1.08	1.28

(Table V continued.)

Year	0	1	2	3	4	5	6	7	8	9
1550	1.70	1.35	1.15	1.23	1.48	1.30	1.03	1.35	1.00	0.98
60	1.20	1.58	1.15	0.98	1.03	1.03	0.93	0.83	0.85	1.10
70	1.20	1.10	1.00	0.88	0.75	1.08	1.25	0.95	0.78	0.88
80	1.00	1.13	0.88	1.13	1.03	0.98	0.88	0.70	0.80	0.78
90	0.73	0.83	0.83	0.68	0.68	0.73	0.70	0.83	0.65	0.53
1600	0.58	0.75	0.68	0.60	0.60	0.88	0.78	0.65	0.60	0.65
10	0.73	0.85	0.90	0.85	1.05	1.03	1.13	1.18	1.00	0.98
20	0.73	1.33	1.00	1.03	1.00	1.25	0.95	1.08	0.85	1.20
30	1.45	1.25	1.30	1.28	0.98	0.93	1.55	1.50	1.68	1.38
40	1.35	1.08	0.90	0.90	0.88	0.93	1.10	1.15	1.05	1.15
1650	1.10	0.90	1.25	1.10	1.13	1.20	1.60	1.48	1.35	1.48
60	1.40	1.55	1.60	1.63	1.34	0.98	1.13	1.28	1.08	1.05
70	0.83	0.78	1.20	0.95	1.00	1.15	0.95	1.13	1.43	1.45
80	1.43	1.30	1.13	1.33	1.33	1.05	1.03	0.95	1.18	1.18
90	1.20	1.28	0.98	0.80	0.85	0.73	0.88	0.88	0.95	0.90
1700	1.45	1.20	1.30	1.35	1.28	1.20	1.23	1.30	1.08	1.35
10	1.40	1.08	0.93	1.13	1.03	1.10	1.03	1.08	1.18	1.00
20	0.90	0.78	1.00	1.00	1.10	0.93	1.08	1.00	0.93	1.05
30	0.95	1.00	0.90	0.88	0.93	0.90	1.00	0.93	1.05	1.05
40	1.20	1.20	0.95	1.10	1.18	1.05	1.15	0.85	0.93	1.05
1750	1.13	1.20	1.30	1.18	1.05	1.03	0.95	1.05	1.08	0.98
60	1.05	1.18	1.30	1.05	0.98	1.05	1.03	0.93	1.10	1.15
70	1.08	0.88	0.83	0.78	0.88	0.93	0.83	0.90	0.95	0.90
80	0.98	1.00	1.03	0.98	0.93	0.85	0.75	0.78	0.85	0.85
90	1.15	0.83	0.95	1.00	1.05	0.90	0.73	1.03	0.90	0.98
1800	1.10	1.13	0.98	1.05	1.10	1.10	1.08	0.90	1.03	1.05
10	0.83	0.73	0.78	0.75	0.85	0.78	0.75	0.80	0.78	0.78
20	0.85	0.98	0.70	0.85	0.73	0.75	0.73	0.55	0.65	0.88
30	0.73	0.73	0.78	0.65	0.70	0.65	0.68	0.78	0.55	0.63
40	0.55	0.73	0.68	0.80	1.05	1.03	0.88	0.83	0.78	0.88
1850	0.75	0.65	0.75	0.85	0.73	0.65	0.75	0.68	0.70	0.73
60	0.65	0.73	0.55	0.60	0.65	0.68	0.80	0.65	0.73	0.90
70	0.95	1.00	0.85	0.98	0.88	0.75	0.78	0.88	1.03	0.90
80	1.18	1.15	1.18	0.90	0.90	0.75	0.88	0.85	0.73	0.78
90	0.83	0.83	0.95	0.80	0.70	0.60	0.80	0.80	0.83	0.85
1900	0.90	0.88	0.93	0.95	0.88	0.90	0.85	1.10	0.98	0.90
10	1.13	1.08	1.00	1.15	0.80	0.95	0.98	0.80	0.80	0.78
20	0.83	—	—	—	—	—	—	—	—	—

If there exist in the different phenomena showing the climatic change, they must be present in the oldest observations available at the present. Since special mention has been made of tree-rings analysis, the related information may be used with the climatic change to establish agreement in their periodicity of the freezing dates. For good reasons this cannot

be done at all, or at best only very imperfectly. In the following, data for the tree-rings analysis in Japan are given for references.

When K. YAMAZAWA (1879–1949) began his studies in tree-rings analysis he was seeking to read and interpret what the trees had been able to record climatic changes. A continuous annual record almost 800 years in length derived from the long-lived Japanese cypress hinoki of Hida prefecture which is located 70–90 km from Lake Suwa, has been established as shown in Table V. It is evident that such records permit an analysis especially for long period rainfall fluctuations, where relatively short meteorological records are powerless. The purpose of this reproduction of Table V is to study the probable relation to freezing dates of Lake Suwa.

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References.

1. FUJIWHARA, S.: Relations among "Omiwatari", sunspots and rich and poor harvests. Geophys. Mag. Tokyo **25**, in press (1954).
2. — Periodic change of climatic elements relating to the oscilation of the Earth's axis. J. Met. Soc. Japan, (1), **31**, 11 (1912).
3. — Notes on the climatic variations concluded from the dates of the first complete freezing of Lake Suwa in Japan. Geograf. Ann. **3**, 358 (1921).
4. TANAKA, A.: Studies on Lake Suwa, Part I. Iwanami Book Co. 1918 (in Japanese), 936 pages.
5. YAMAZAWA, K.: Japanese Climatic Chronology, Sumii Book Co. 1936 (in Japanese), 64 pages.